Module-2

Machine tool Operations

Introduction:

Manufacturing process may be defined as a process of converting, the raw material into your finished product of required size and shape depending upon the various processes and methods. Manufacturing process is broadly classified into two groups.

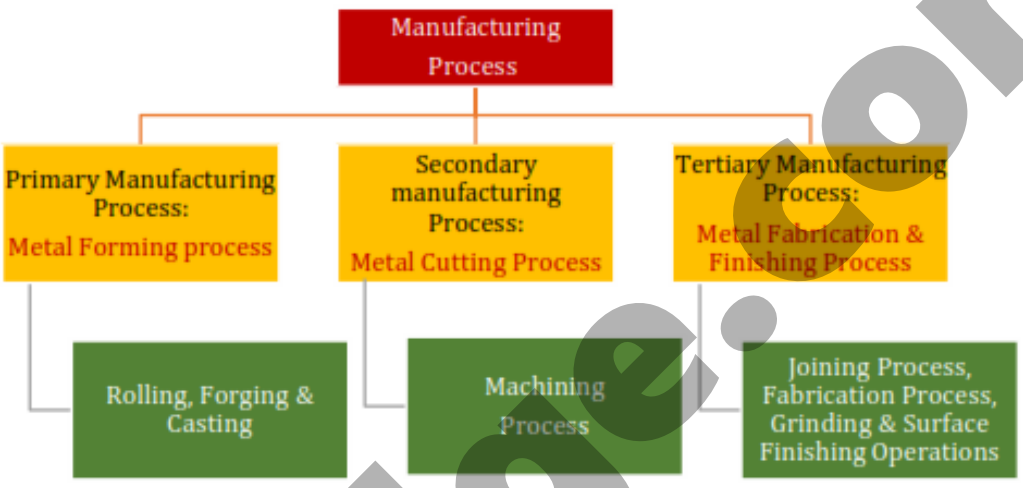


Figure: Block diagram of Classification of manufacturing Process

In metal cutting process, extra amount of material is removed from the workpiece in the form of chips. The material removed due to the relative motion between the cutting tool and the work-piece.

In metal forming process, no material is removed, but it is forced to change its size and shape by applying external pressure.

Machine Tools:

When machines perform the metal cutting operations by the cutting tools mounted on them, they are called “Machine tools”. A machine tool may be defined as a power-driven machine which accomplishes the cutting or machining operations on it. The machine tools used for most of the machining processes are Lathe, Drilling, Planing, Milling and Grinding machines

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Lathe Machine Tool:

The Lathe is one of the oldest machine tools used to produce cylindrical objects. The modern engine late are first developed in the year 1797 by an Englishman Henry Maudslay. The main function of the lathe is to removal of metal from a work piece of work to give it a required shape and size. This is accomplished by holding the work securely and rigidly on the machine and then turning it against a cutting tool which will remove the material from the work in the form of a chip. To cut the material properly, the tool should be harder than that of the material of the workpiece.

Principle working of a Lathe Machine Tool:

Lathe machine is one of the most important machine tools which is used in the metalworking industry. It operates on the principle of a rotating work piece and a fixed cutting tool. The cutting tool is feed into the work piece which rotates about its own axis causing the workpiece to form the desired shape.

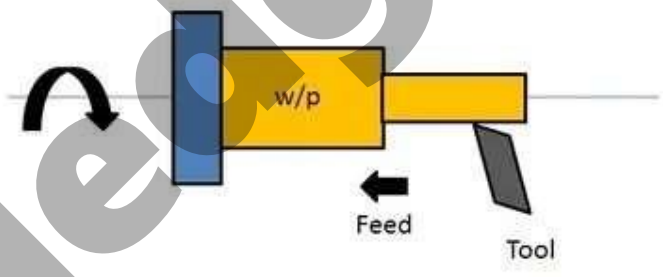


Figure: Principle of turning operation on Lathe

Classification of Lathe Machine Tools:

Lathes are classified based on their design, construction and application. Following are some of the important lathes used in industries.

1. Speed Lathe
2. Engine Lathe
3. Bench Lathe
4. Tool Room Lathe
5. Capstan & Turret Lathe
6. Special Purpose Lathe
7. Automatic Lathe
8. CNC Lathe

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Construction & Working Principle of Centre Lathe/ Engine Lathe:

The schematic arrangement of the Centre or Engine Lathe is as shown in the following figure. The major parts of a centre lathes are:

1. Bed & Guideways
2. Headstock
3. Tailstock
4. Lathe bed
5. Carriage Assembly
6. Driving mechanism & Feeding mechanism

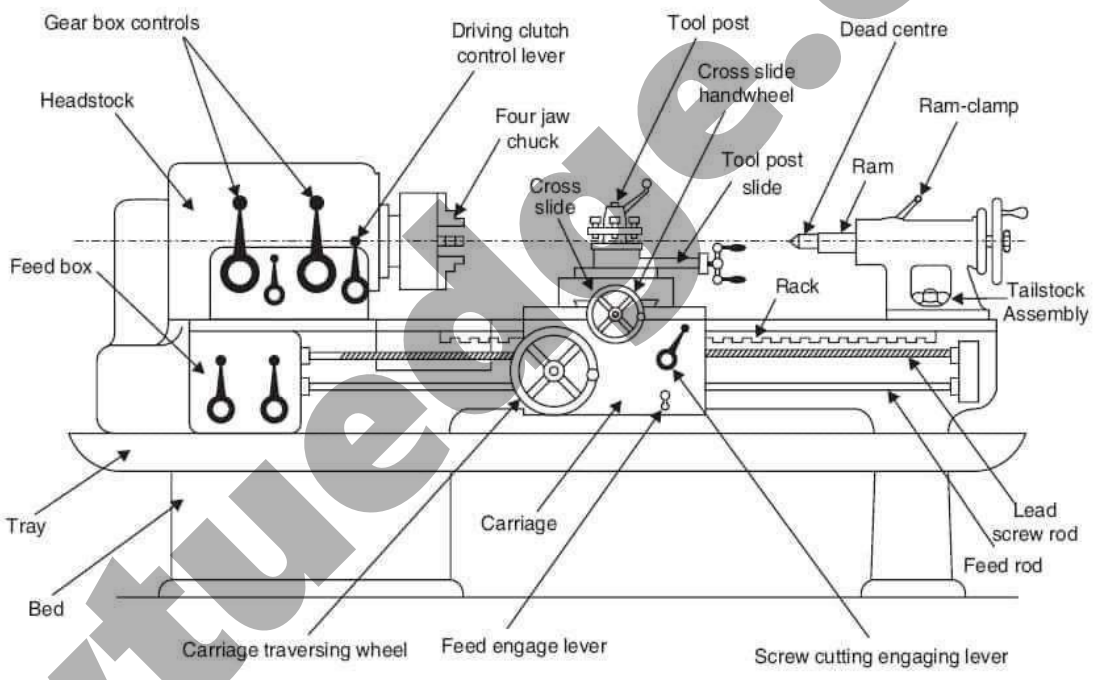


Figure: Schematic arrangement of Centre Lathe.

Bed: The bed forms, the base of the machine. The headstock and the tail stocks are located at either end of the bed and the carriage rest over the lathe bed and slides on it. The guide ways of the lathe bed may be flat and inverted V type having an include angle of 90 degree.

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Headstock: The headstock is secured permanently on the inner waves, at the left end of the lathe bed and it provides mechanical means of rotating the work at multiple speeds, It contains a hallow spindle and mechanism for driving and altering this spindle speed. All the parts are housed within the headstock casting.

Tailstock: The tail stock is located on the inner waves at the right end and top of the bed ways. This has two main uses.

- It supports the other end of the workpiece.
- It holds the tool for performing operations, such as drilling. Riming tapping etc.,

The body of the Tailstock can be adjusted along the length of the bed-ways by sliding it to the desired position, and can be locked on a lathe bed-way.

Carriage: The carriage of a lathe has several parts that serves to support, move and control the cutting tool. It consists of the following parts saddle cross, slide, Compound slide, tool post and apron.

- **Saddle:** The saddle is an H- shape casting that fits over the bed and slides along the base. It carries the cross slide and tool post.
- **Cross slide:** It is mounted on the saddle and enables the movement of the cutting tool laterally, across the lathe bed by means of a cross feed handle. It also serves as the support for the compound rest.
- **Compound rest:** The compound rest is mounted on top of the cross slide and has a circular base graduated in degrees. It is used for obtaining angular cuts and short tapers. As well as convenient positioning of the tool at the work.
- **The tool post:** This is located on top of the compound slide to hold the tool and enable it to be adjusted to convenient Working position,

Feed mechanism: The movement of the tool related to the work is called as a feed. The feed can be given either by the hand, or by automatic or powered feed. To get the automatic feed, carriage is engaged to feed rod. while for cutting threads, carriage is engaged to screw rod.

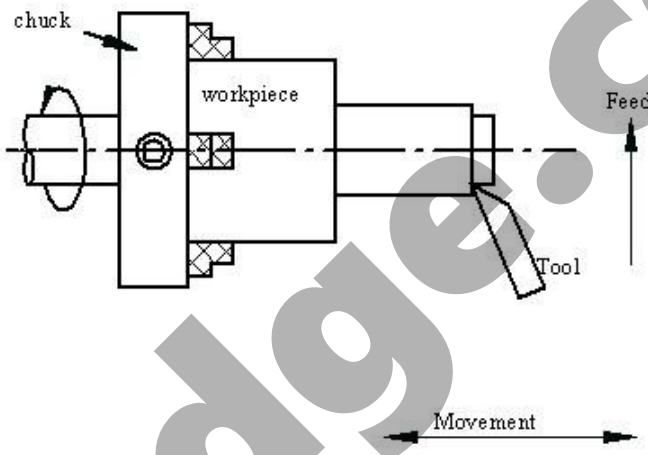
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Lathe Operations:

Some of the operations carried out on a lathe machine are as listed below;

1. Facing
2. Turning
3. Step turning
4. Taper turning
5. Knurling
6. Thread cutting
7. Drilling
8. Counter sinking
9. Contour forming
10. Boring etc.,

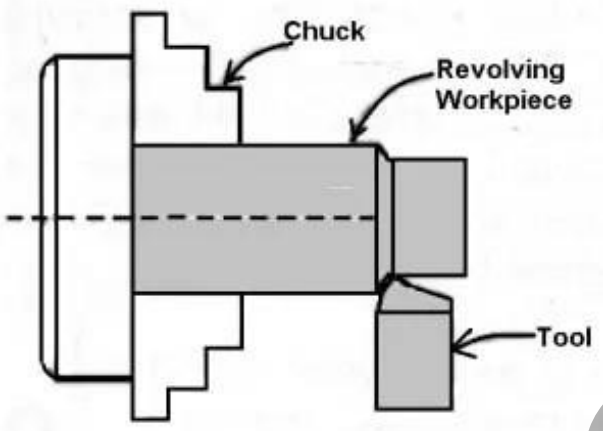
1. Facing:



It Is the operation of machining the ends of a piece of the work to produce a flat surface perpendicular with the axis. This is also used to cut the work to the required length. The operation involves feeding the tool perpendicular to the axis of rotation of the work piece. A regular cutting tool may be used for facing a large work piece. The cutting edge should be set at the same height as the centre of the work piece. A properly ground facing tool is mounted in a tool holder in the tool post to accomplish facing operation.

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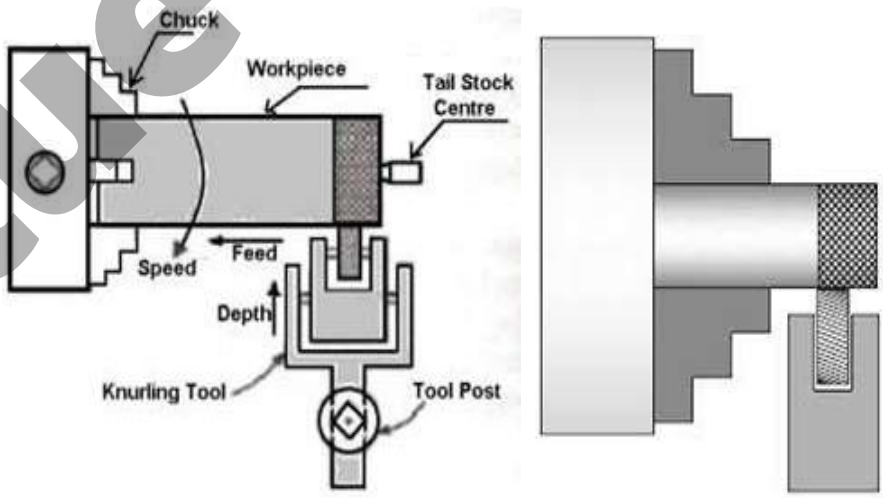
2. Turning Operation:



The process of metal removal from the cylindrical jobs is called straight or plain turning. Cross-slide and the carriage are used to perform turning operation and make the operation faster and economical. Plain turning operations are generally performed in two steps-rough and finish turning.

Rough turning is usually done for rolled, cast or forged parts to remove the uneven or sandy or rough surface on the jobs. A roughing tool does roughing and used for excess stock removal. For finishing a tool with slightly round cutting edge is used. The depth of cut rate is at the range of 0.2 to 1 mm and the feed rate between 0.1 to 0.3 mm.

3. Knurling:



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Knurling is an operation performed on the lathe to generate serrated surface on the work piece. This is used to produce a rough surface for griping like the barrel of the micrometre or screw gauge. This is done by a special tool called knurling tool which has a set of hardened rollers with the desired serrations.

During knurling operation, the hardened rollers of the tool are pressed against the slowly rotating work pieces such that the impression of tool serrations are formed on the work pieces surface.

4. Taper turning:

Taper turning is an operation to produce conical surface on the work piece. This can be machined by either work piece inclined to the axis of the lathe or tool moving inclined to the axis of the lathe.

There are many methods for taper turning

a) By swivelling the compound rest.

b) By offsetting the tail stock.

c) By taper turning attachment.

d) By using form tool/ Broad nose tool

e) Combination of Longitudinal and Transverse feed movement.

a) Taper Turning by Swiveling the compound Tool Rest

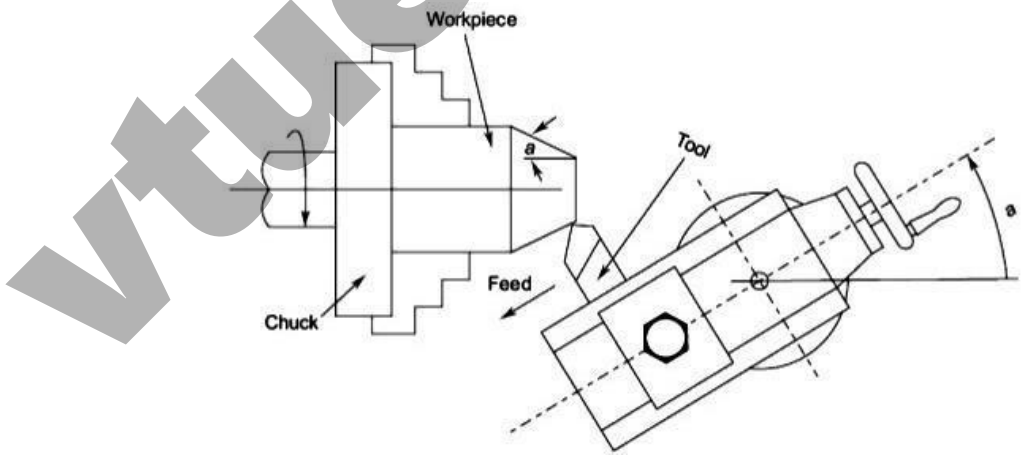


Figure: Taper turning by Swiveling the compound rest method

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The work pieces which require steep taper for short lengths the taper turning was done by swiveling the compound rest method. In this method the compound tool rest is swiveled to the required taper angle and then locked in the angular position. The carriage is also locked at that position. For taper turning the compound rest is moved linearly at an angle so that the cutting tool produces the tapered surface on the work piece. This method is limited to short taper lengths due to the limited movement of the compound tool rest.

The taper angle is calculated by,

$$\tan\alpha = \frac{(D-d)}{2l}$$

Where,

D= bigger diameter of the taper (mm)

d= smaller diameter of the taper (mm)

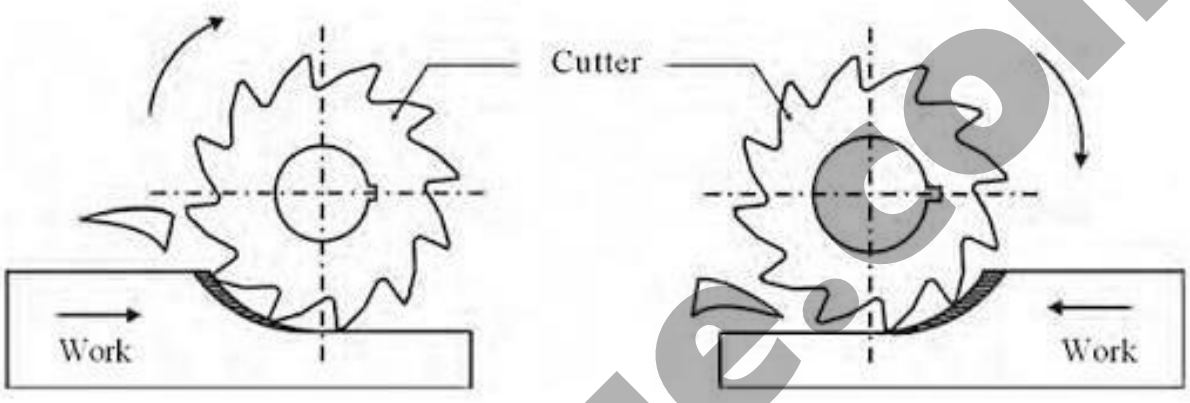
L= Length of the taper (mm)

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Milling Machine Tool

Milling is a metal cutting operation in which the cutting tool is a slow revolving cutter having cutting teeth formed on its periphery. The milling cutter is a multipoint cutting tool. The work piece is mounted on a movable worktable, which will be fed against the revolving milling cutter to perform the cutting operation.

Principle of Milling



(a) (b)

Figure: (a) Up milling and (b) Down milling operation

Figure shown above is the principle of cutting action of a milling cutter. The milling cutter is mounted on a rotating shaft known as arbor. The work piece which is mounted on the table can be fed either in the direction opposite to that of the rotating cutter as shown in above fig (a) or in the same direction to that of the cutter as shown in above fig (b). When the work piece is fed in the opposite direction to the cutter tooth at the point of contact, the process is called as conventional or up-milling.

Milling Machine Tool: A Milling Machine is a machine tool in which a stationary work piece is fed against a rotating multi point cutter.

Various types of milling machines are

1) Plain or horizontal type of milling machine.

2) Vertical Milling Machine

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Horizontal Milling Machine

The schematic arrangement of horizontal milling machine/ column and knee typr milling machine is as shown in the figure.

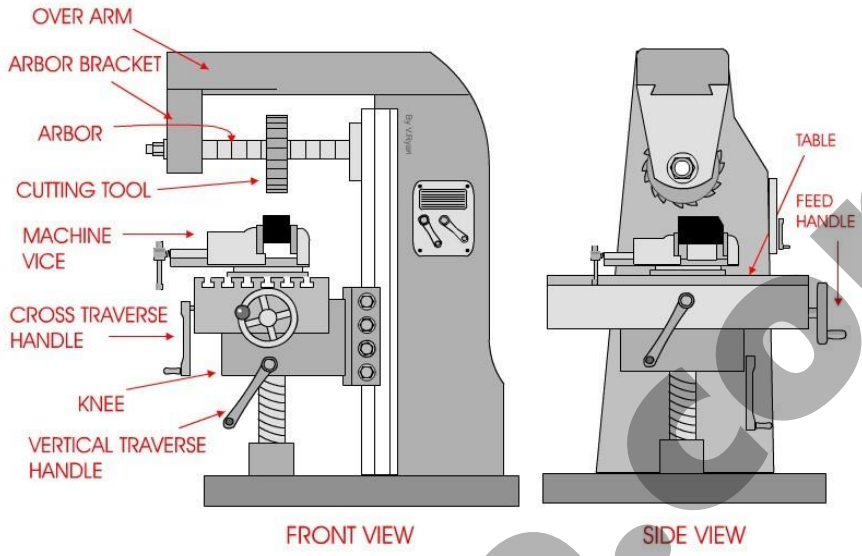


Figure: Horizontal / Column & Nee type milling machine

Milling machine has some principal parts. Their understanding would help us in understanding how the machine operates. The various parts are

1. Base
2. Column
3. Arbor
4. Knee
5. Saddle
6. Table
7. Overhanging arm
8. Driving and feeding Mechanism

Base: Base forms the foundation of the machine tool. It's a rectangular casting made up of Cast Iron.

Column: The column is another rectangular casting mounted on one end of the base. The column is ribbed heavily in order to support the knee.

Arbor: The arbor is a horizontal shaft provided with a straight body and tapered shank. On the straight portion of arbor, rotary cutters are mounted.

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Saddle: The saddle is casting provided with two slides one at the top and the other at the bottom, which are exactly right angles to each other. The lower slide fits within the guideways on the top of the knee and the upper slide receives the dovetail guides provided on the bottom of the table.

Table: The table is mounted on the top of the saddle. The bottom of the table has a dovetail slide which fits in the slide way on the top of the saddle. The top the table is machined with full length T-slots for mounting vices or other work holding fixtures.

Overhanging Arm: The overhanging arm extends from the column in order to support the arbor. The front brace attached to the overhanging arm supports the free end of the arbor and thus provides rigidity to the rotating cutter.

Milling machine Operations/ Applications:

A variety of milling operations are performed on a milling machine to produce a horizontal, vertical, inclined surfaces, keyways, slots, gear teeth etc., A few of the most commonly used milling operations are described here.

1. Slab Milling

This method is also called as plain milling. The plain milling is the most common types of milling machine operations. Slab milling is performed to produce a plain, flat, horizontal surface parallel to the axis of rotation of a plain milling cutter.

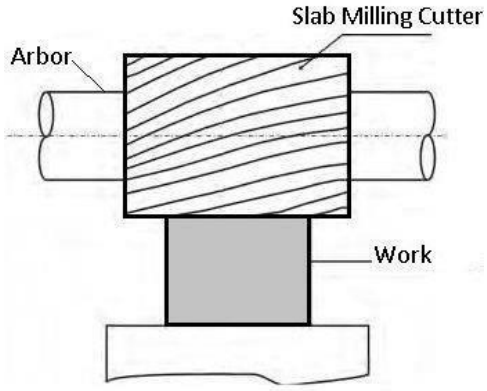
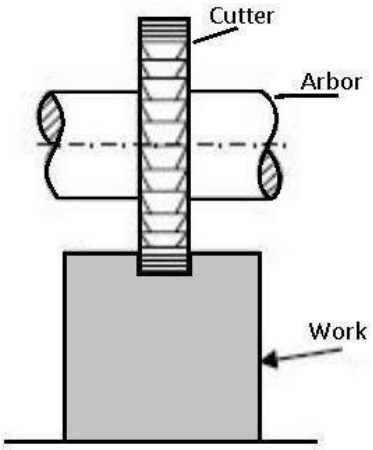
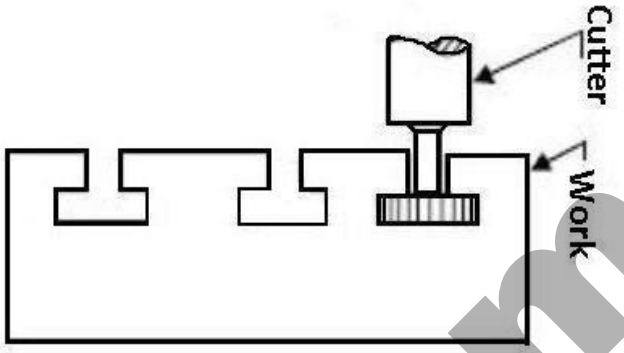


Figure: Slab/ Plain Milling Operation

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2. Slot Milling





The slot milling, also called as groove or keyway cutting is done on a horizontal milling cutter or using an end milling cutter.

- The operation of producing of keyways, grooves and slots of varying shapes and sizes can be performed in a milling machine.
- It is done by using a plain milling cutter, a metal slitting saw, an end mill or by a side milling cutter.
- The open slots can be cut by a plain milling cutter, a metal slitting saw, or by a side milling cutter. The closed slots are produced by using endmills.

3. Face Milling

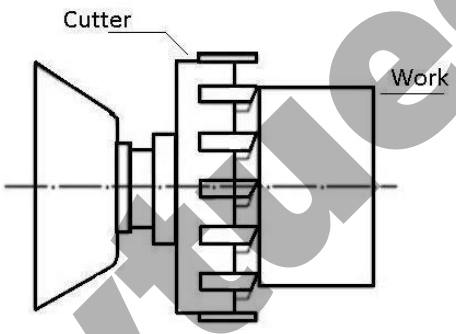


Figure: Face Milling operation

In this method, the flat surfaces are milled by the cutters, which are mounted with their axes perpendicular to the milled surfaces.

- The face milling is the simplest milling machine operations.
- This operation is performed by a face milling cutter rotated about an axis perpendicular to the work surface.

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- The operation is carried in plain milling, and the cutter is mounted on a stub arbor to design a flat surface.
- The depth of cut is adjusted by rotating the cross-feed screw of the table.

Drilling Machine Tool

Introduction

Drilling machine is one of the most important machine tools in a workshop. It was designed to produce a cylindrical hole of required diameter and depth on metal workpieces. Though holes can be made by different machine tools in a shop, drilling machine is designed specifically to perform the operation of drilling and similar operations. Drilling can be done easily at a low cost in a shorter period of time in a drilling machine.

Types of drilling machines

Drilling machines are manufactured in different types and sizes according to the type of operation, amount of feed, depth of cut, spindle speeds, method of spindle movement and the required accuracy.

The different types of drilling machines are:

1. Portable drilling machine (or) Hand drilling machine
2. Sensitive drilling machine (or) Bench drilling machine
3. Upright drilling machine
4. Radial drilling machine
5. Gang drilling machine

Upright drilling machine

The upright drilling machine is designed for handling medium sized workpieces. Though it looks like a sensitive drilling machine, it is larger and heavier than a sensitive drilling machine. Holes of diameter up to 50mm can be made with this type of machine. Besides, it is supplied with power feed arrangement. For drilling different types of work, the machine is provided with a number of spindle speeds and feed.

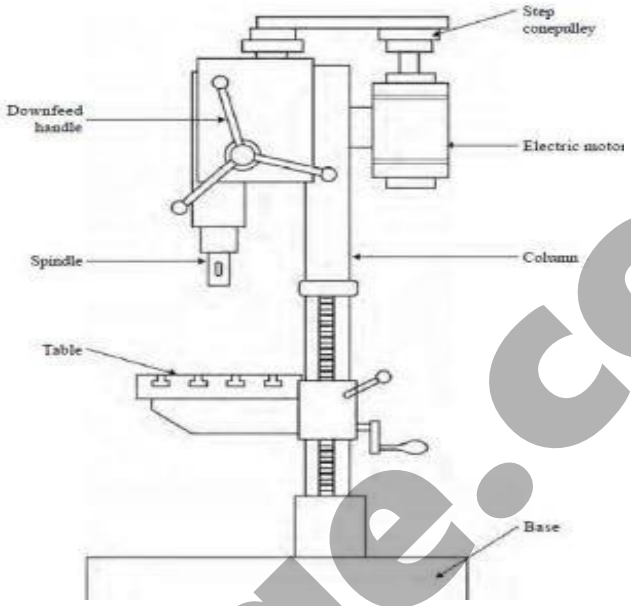
Base

The base is a rectangular casting made of cast iron and so can withstand vibrations. It is mounted on a on the floor. It supports all the other parts of the machine on it.

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Column

The column stands vertically on the base at one end. It supports the work table and the drill head. The drill head has drill spindle and the driving motor on either side of the column.



Table

The table is mounted on the vertical column and can be adjusted up and down on it. The table has 'T'-slots on it for holding the workpieces or to hold any other work holding device.

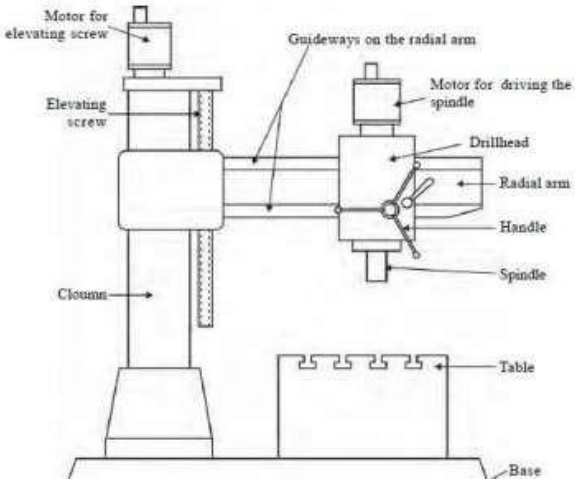
Drill head

Drill head is mounted on the top side of the column. The drill spindle and the driving motor are connected by means of a V-belt and cone pulleys. The motion is transmitted to the spindle from the motor by the belt.

Radial drilling machine

The radial drilling machine is intended for drilling on medium to large and heavy workpieces. It has a heavy round column mounted on a large base. The column supports a radial arm, hich can be raised or lowered to enable the table to accommodate workpieces of different heights. The arm, which has the drill head on it, can be swung around to any position. The drill head can be made to slide on the radial arm. The machine is named so because of this reason. It consists of parts like base, column, radial arm, drill head and driving mechanism. A radial drilling machine is illustrated in Figure.

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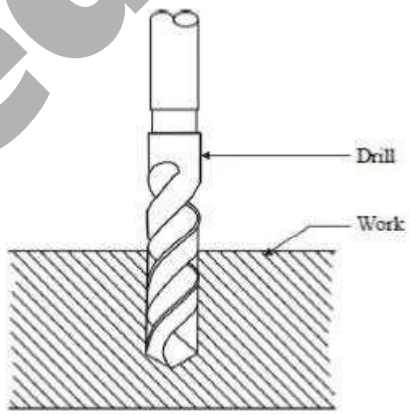


Drilling machine operations

Though drilling is the primary operation performed in a drilling machine, a number of similar operations are also performed on holes using different tools. The different operations that can be performed in a drilling machine are:

1. Drilling	5. Countersinking
2. Reaming	6. Spot facing
3. Boring	7. Tapping
4. Counter boring	8. Trepanning

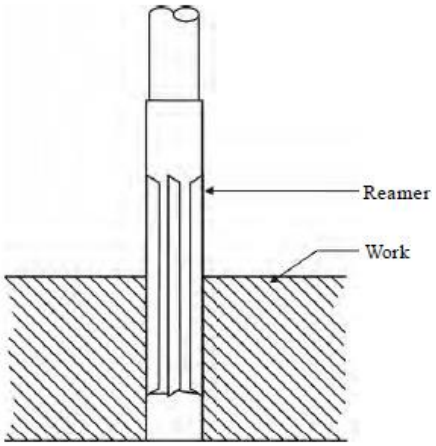
1. Drilling:



Drilling can be called as the operation of producing a cylindrical hole of required diameter and depth by removing metal by the rotating edges of a drill. The cutting tool known as drill is fitted into the spindle of the drilling machine. A mark of indentation is made at the required location with a centre punch. The rotating drill is pressed at the location and is fed into the work. The hole can be made up to a required depth.

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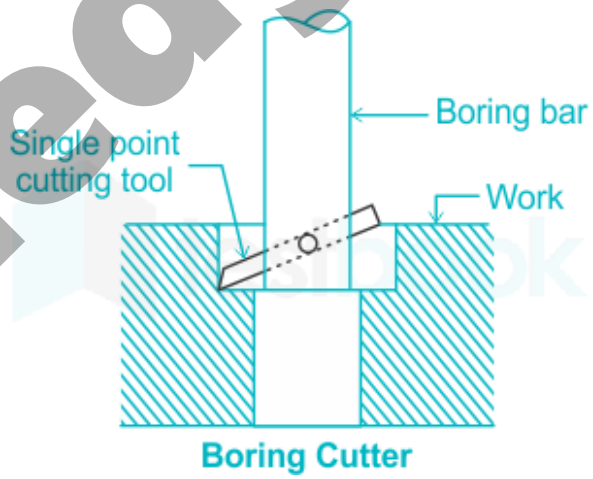
2. **Reaming:**



Reaming

Reaming shown in Figure, is an accurate way of sizing and finishing a hole which has been previously drilled. In order to finish a hole and to bring it to the accurate size, the hole is drilled slightly undersize. The speed of the spindle is made half that of drilling and automatic feed may be employed. The tool used for reaming is known as the reamer which has multiple cutting edges. Reamer cannot be used to drill a hole. It simply follows the path which has been previously drilled and removes a very small amount of metal.

2. Boring

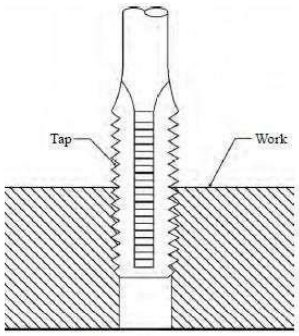


Boring Cutter

Boring is the operation of enlarging the size of the previously drilled hole. For this purpose a special purpose cutting tool is used. Boring operation is also carried out to finish a hole accurately and to bring it to the required size.

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4. Tapping:



Tapping shown in Figure is the operation of cutting internal threads by means of a cutting tool called a tap. A tap may be considered as a bolt with accurate threads cut on it. The threads act as cutting edges which are hardened and ground. When the tap is screwed into the hole it removes metal and cuts internal threads which will fit into external threads of the same size.

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Introduction to Modern Manufacturing Tools and Techniques

Computer Numeric Control [CNC] Machine:

Introduction: Computer Numeric Control (CNC) is the automation of machine tools that are operated by precisely programmed commands encoded on a storage medium (computer command module, usually located on the device. Most NC today is computer (or computerized) numerical control (CNC), in which computers play an integral part of the control.

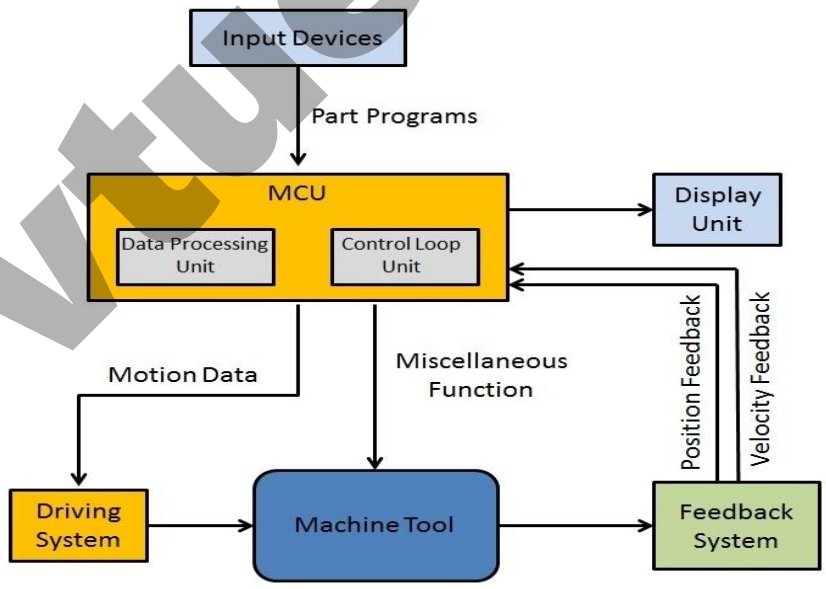
Definition

Computer Numerical Control (CNC) is one in which the functions and motions of a machine tool are controlled by means of a prepared program containing coded alphanumeric data. CNC can control the motions of the work piece or tool, the input parameters such as feed, depth of cut, speed, and the functions such as turning spindle on/off, turning coolant on/off etc,.

Components of CNC:

A CNC machine tool following units in its construction,

1) Input device
2) MCU or machine control unit
3) Machine tool
4) Driving System
5) Feedback devices
6) Display Unit



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1. **Input Devices:** In put devises are the devises used to enter the part program into the machine tool. Universal Serial Bus (USB), ethernet communication, Tapes and recording devices can be used as input devices.
2. **Machine Control** Unit: The machine control unit (MCU) is a microcomputer that stores the program and executes the commands into actions by the machine tool. The MCU consists of two main units: the data processing unit (DPU) and the control loops unit (CLU). The DPU processes the data from the part program and provides it to the CLU which operates the drives attached to the machine leadscrews and receives feedback signals on the actual position and velocity of each one of the axes. A driver (dc motor) and a feedback device are attached to the leadscrew. The CLU consists of the circuits for position and velocity control loops, deceleration and backlash take up, function controls such as spindle on/off.
3. **Machine Tool:** The machine tool could be one of the following: lathe, milling machine, laser, plasma, Coordinate measuring machine etc. Machine tool actuated as per the program through various mechanical driving and feeding mechanisms. Machine tool perform the intended task over the work piece.
4. **Driving system:** A drive system essentially is made up of amplifier circuits, drive motors and ball lead screws. Commonly used types of electrical motors include DC Servo motors, AC servo motors, Stepping motors and Linear motors
5. **Feed back devices:** For accurate operation of a CNC machine, the positional values and speed of the axes need to continuously updated. Inorder to perform this task Positional feed back devices and velocity feed back devices are used.

   The positional feedback systems gives the position of the tool/ workpice with respect to the reference point. Velocity feed back system gives the state of the velocity of the sped of the spindle, cutting tool and the movement of the table etc.,
6. **Display Unit:** Device that ensures interaction between the machine operator and the machine is called as a display unit. It displays the current status of the operation such as the spindle RPM, running part program, feed rate, position of the machine slide etc. it also shows the graphic simulation of the path taken by the tool and that the operator can verify the part program before actual machining and any malfunction of the CNC system is also displayed as warnings.

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Advantages of the CNC Machine Tool:

Advantages:

•High Repeatability and Precision e.g. Aircraft parts

•Volume of production is very high

•Complex contours/surfaces need to be machined. etc

•Flexibility in job change, automatic tool settings, less scrap

•More safe, higher productivity, better quality

•Less paper work, faster prototype production, reduction in lead times

Disadvantages of the CNC Machine:

Disadvantages:

•Costly setup, skilled operators

•Computers, programming knowledge required

•Maintenance is difficult

CNC Machining Centres:

- The term "machining centers" describes almost any CNC milling and drilling machine that includes an automatic tool changer and a table that clamps the workpiece in place.
- CNC machine center is a advance manufacturing machine tool which performs wide range of machining operation with accuracy and good quality surface finish.
- The orientation of the spindle is the most fundamental defining characteristic of a CNC
- machining centers.
- CNC machining centers can further be classified based on the rotation of either the work piece Or the rotation of the tool as:

1. CNC turning machines
2. CNC Milling machines

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CNC Turing Centres:



- The primary function of a CNC Turning Center is that it rotates (or "turns") your workpiece.
- CNC Turning Machines are one of the oldest and simplest forms of machining parts, called "lathes," .
- Can be either horizontal or vertical depending on the weight and tolerance of the workpiece.
- Work-peices for this process are usually round, but can be other shapes — like squares or hexagons.
- The work-piece is held in place by an instrument known as the "chuck." The chuck then spins at various RPMs (depending on the capability of your machine).
- When this occurs, the machine's tool moves into the rotating workpiece and begins to shave away material to create the desired shape.

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CNC Milling Centres/ Machining Centres:



- The primary function of a CNC Milling Machine is that your tool will be doing the rotating and moving while your workpiece stays in one spot (generally).
- Milling is a more specific process that is similar to drilling and cutting.
- These machines can also be either horizontal or vertical, again depending on the tolerance and weight of your workpiece.
- This process has many axes that allow for a variety of shapes, holes, and slots to be cut into the workpiece at many angles.
- These axes provide many different maneuvers, either by the spindle or the bed, to cut the part desired to the exact specifications.

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